

Economic Analysis of Cleaner Industrial Boilers in Nepal: Fuel Choices, Efficiency, and Policy Implications

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Outline

- Introduction
- Methodology and Data
- Results
- Conclusions



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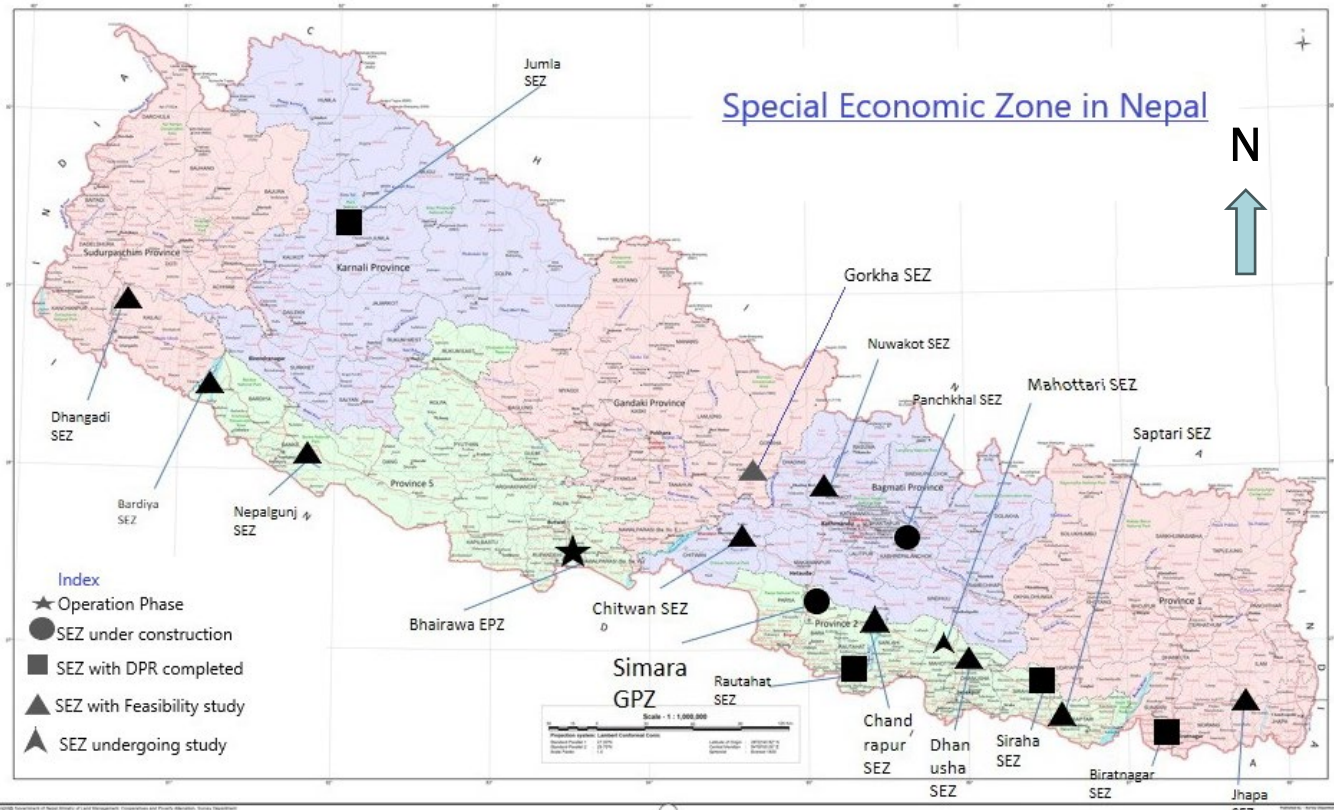
Manufacturing sector profile

No. of establishments*	4,076 (2012)	5,017 (2020)
Food and beverages	26%	25%
Non-metallic minerals	23%	22%
Textiles	9%	9%
Fabricated metals	7%	9%

	2015	2024
Contribution to GDP	5.9%	5.3%
AAGR of sector	2.3%	

Source: MoF (2024)

Sources: CBS (2014, 2022). * Large manufacturing industries (10 or more people)



Introduction (1/2)

- Thermal applications use **two-thirds** of the manufacturing sector's total final energy use. ~ **15%** of the nation's **total final energy consumption**.
- ~ **83%** fuels used in **boilers** come from domestic **biomass**, **14%** from imported **coal and oils**, and the remaining **3%** from **electricity**.
- **Higher thermal energy intensity** translates into **increased energy costs**.
- ~ **20%** of **PM2.5** emissions in Kathmandu and the Terai region are attributable to industrial thermal applications (WB study).
- ~ **15 thousand** deaths (2019 data) attributed to **ambient** (outdoor) air pollution (WHO study).

Introduction (2/2)

- **Switching** to cleaner boiler technologies could provide **economic, health, and environmental benefits** for the country.
- **First**, we analyze the economics of various fuel/technology alternatives for boilers used in industries from private and social perspectives.
- **Second**, we examine incentive designs that promote the adoption of cleaner boilers.

Outline

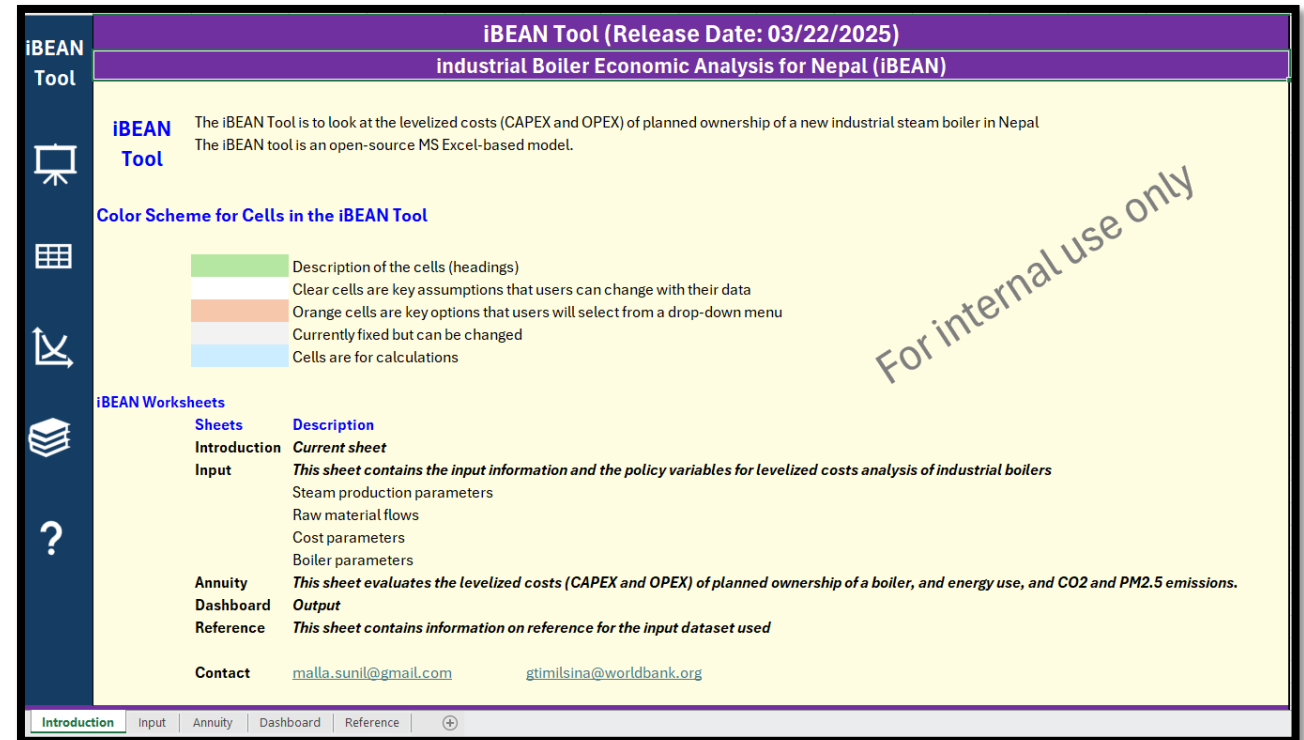
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Methodology (1/2)

We developed an open-source Excel-based iBEAN tool.

- Industrial Boiler Economic Analysis for Nepal (iBEAN)
- **Estimates** the annual and total levelized costs of a new industrial steam boiler.
- **Identifies** relevant incentive designs to promote cleaner boilers.



Methodology (2/2)

Levelized Cost Analysis

$$LCOB = CAPEX + OPEX^* + FEX$$

Where,

LCOB is the levelized cost (annuity) of boiler;

CAPEX is capital cost calculated using $CRF = \frac{i(1+i)^n}{(1+i)^n - 1}$

OPEX* is annual operating expenditure (labor and others but exclude fuel cost);

FEX is fuel cost estimated using $F_p \left(\frac{1}{CV_f} \times \frac{UE}{\eta_b} \right)$, where, **F_p** is fuel price, **CV_f** is gross calorific value of fuel, **UE** is useful energy required per year; **η_b** is rated boiler efficiency.

Data (1/2)

Parameters and variables required

Technical

- Capacity
- Lifetime
- Operational factors
- Efficiency
- Calorific value
- Feedstock requirement
- Labor requirement

Cost

- **CAPEX**
 - Purchase price
 - Market interest rate
 - Loan period
- **OPEX**
 - *Variable costs*
 - Feedstock price
 - O&M
 - *Fixed costs*
 - Labor
 - Installation
 - Insurance/Contingency

Economic/Financial

- Interest rate
- Discount rate
- Down payment
- Equipment grant
- Inflation

Data were collected through primary (survey) and secondary sources.

Data (2/2)

Technical (heat value)

Cost data (fuel price)

	(kCal/kg)	Fuel price*
Coal	5,500	40
Diesel	10,800	181
Fuel oil	10,500	113
Bagasse	4,400	8
Wood	3,500	11
Husk	3,000	13
Pellet	4,000	33
Electric	860	9
LPG	11,307	135

Economic data

Plant Lifetime	20
Discount rate	7%
Loan term (year)	10
Interest rate	8%

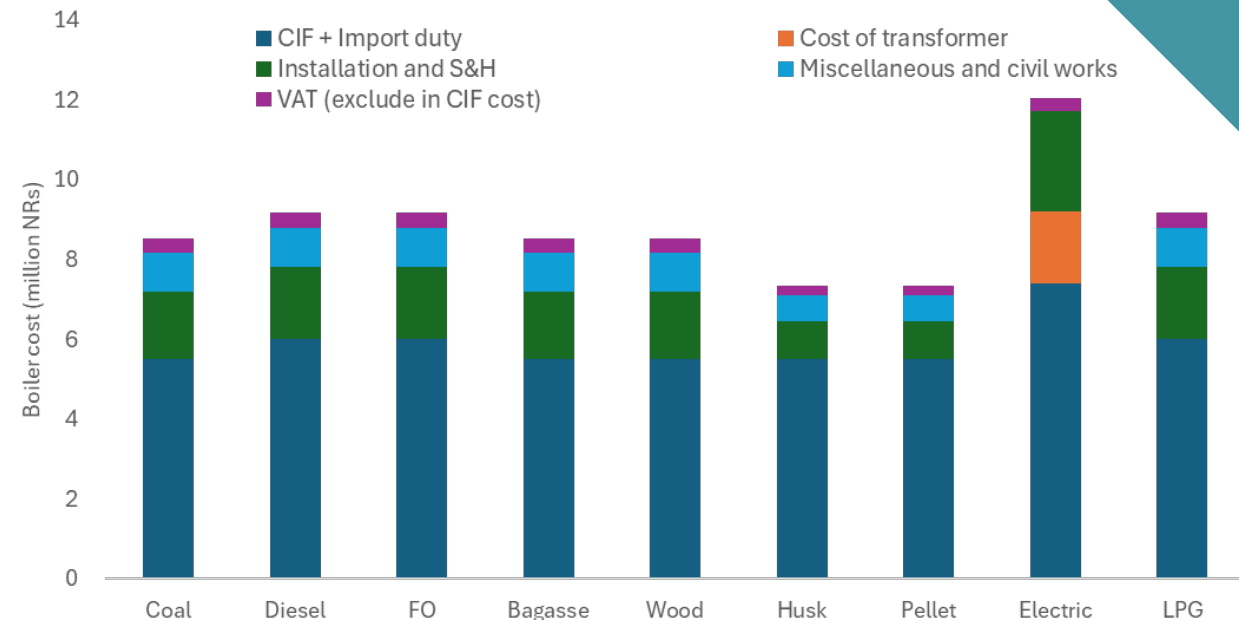
Exchange rate

1 US\$	140 NRs
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* Electricity is in NRs/kWh, others are in NRs/kg.

Plant capacity (ton/hour)	1
No. of operation (hour/year)	3,600
Thermal efficiency	79% - 99%

Equipment cost for 1 tph capacity (million NRs)

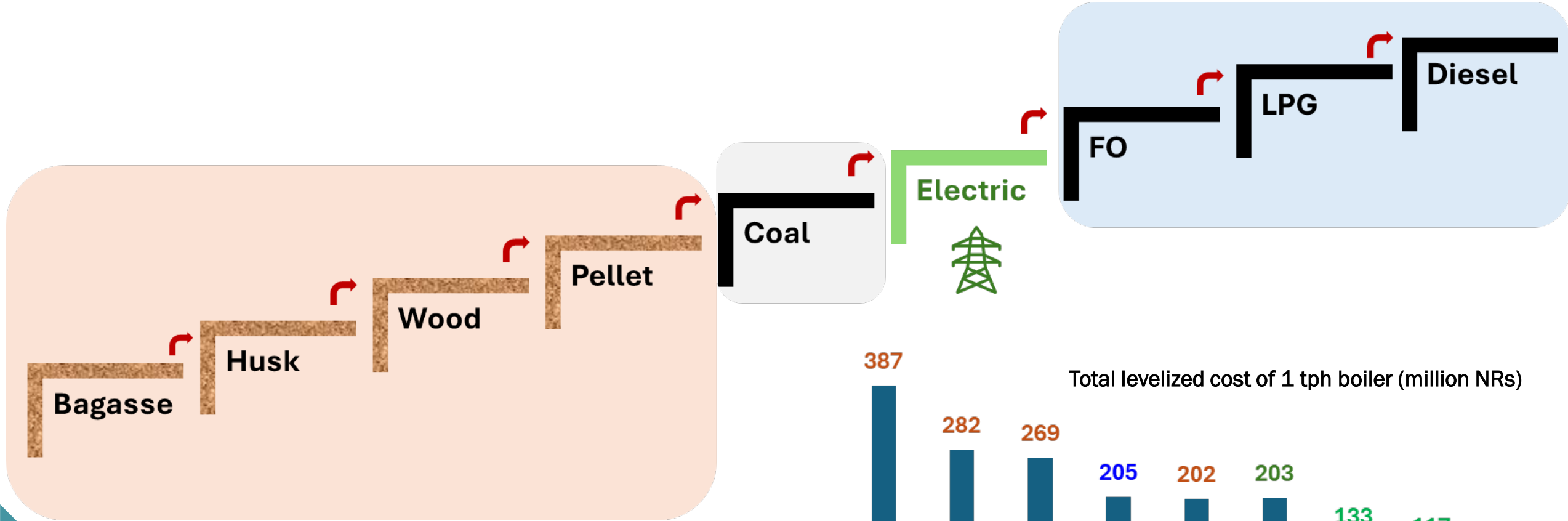


- Purchase of an **electric** boiler is currently the **most expensive**.
- **Other boilers** cost around **72% to 90%** of the electric boiler.

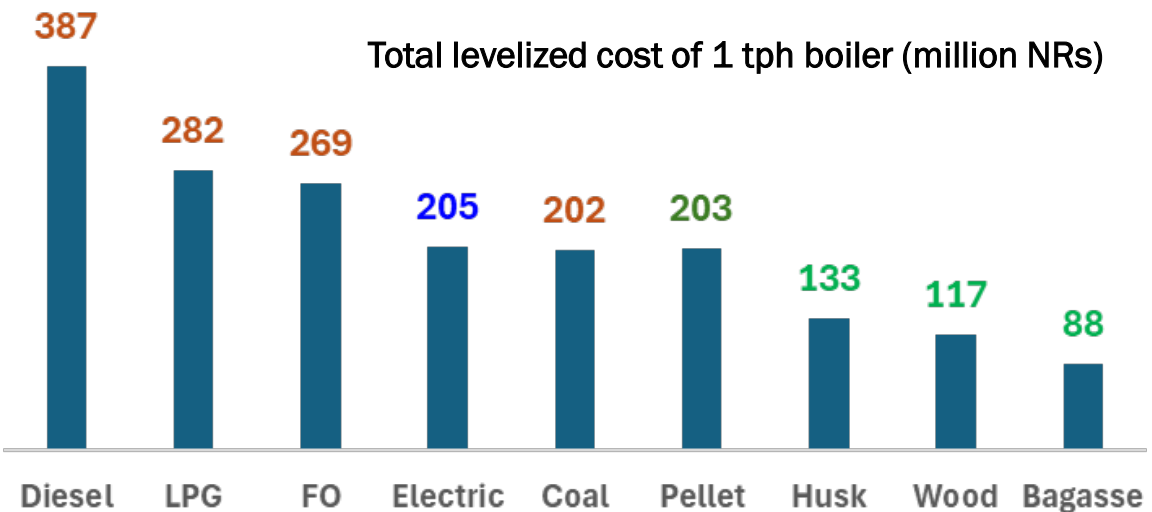
Outline

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- **Results**
 - Base case
 - Alternative scenarios
- Conclusions

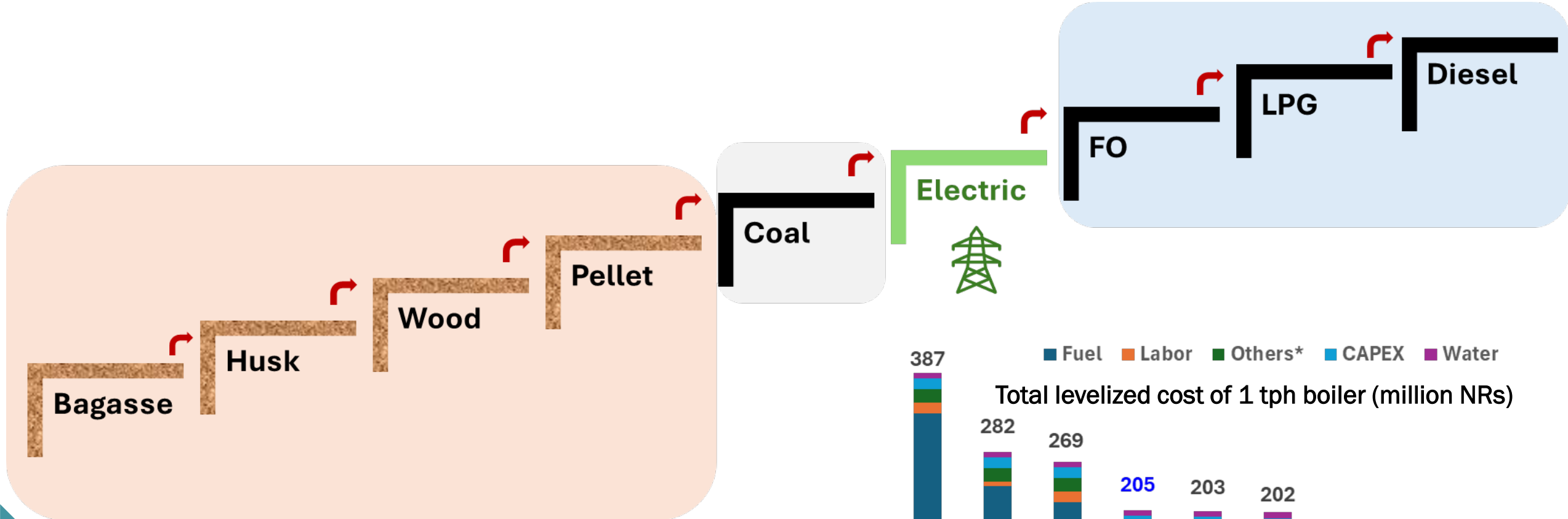




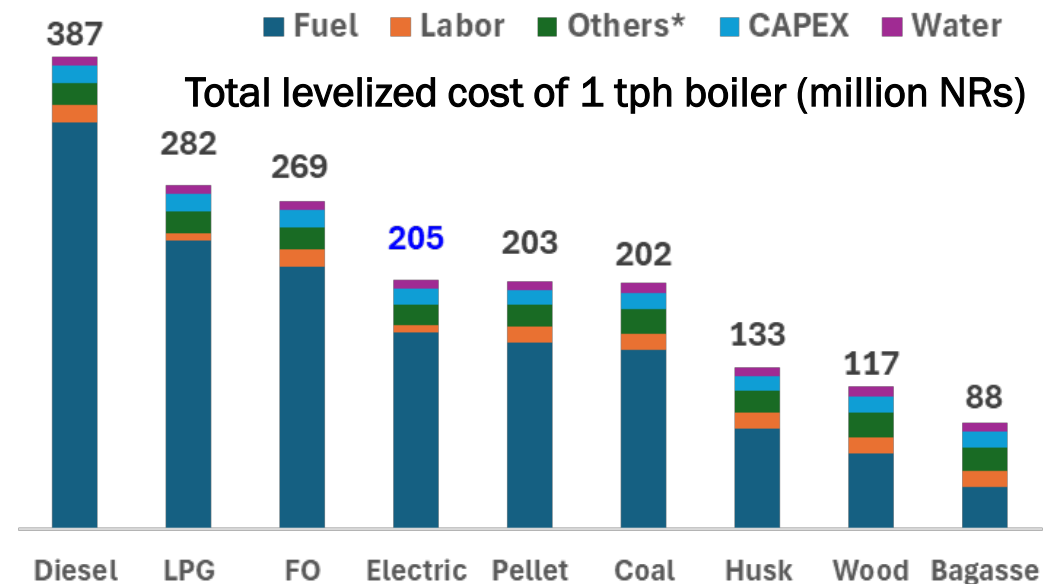
Electric boilers are cheaper than diesel, LPG, and fuel oil but much more expensive than coal and biomass boilers.



Total Levelized cost (Base case)

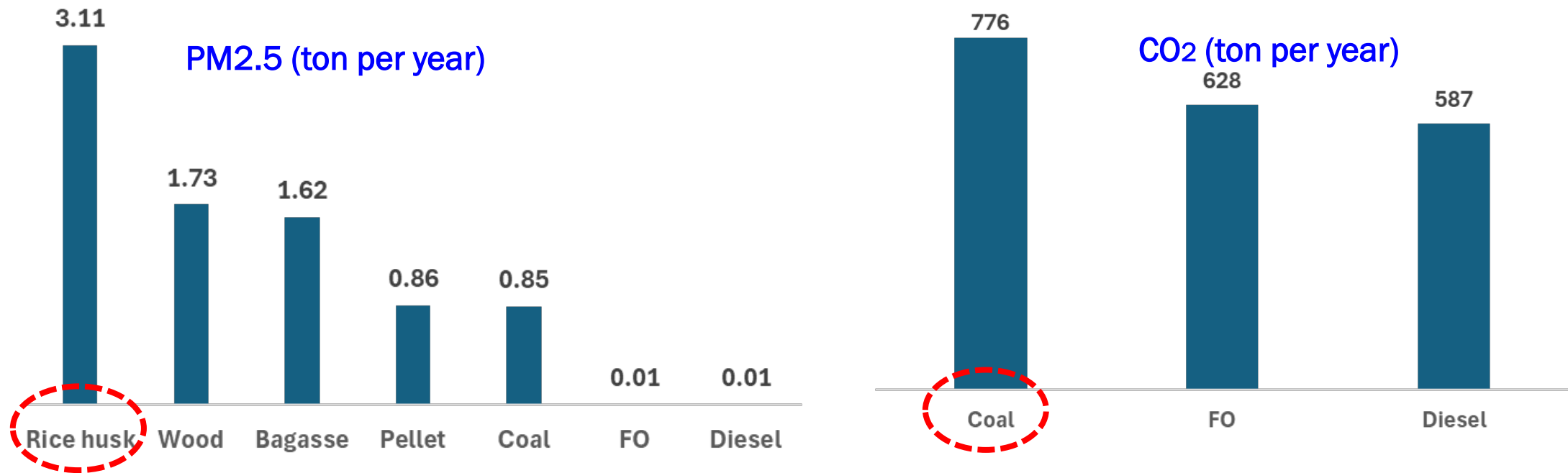


Fuel cost is the major component of boiler cost (39% - 86%)



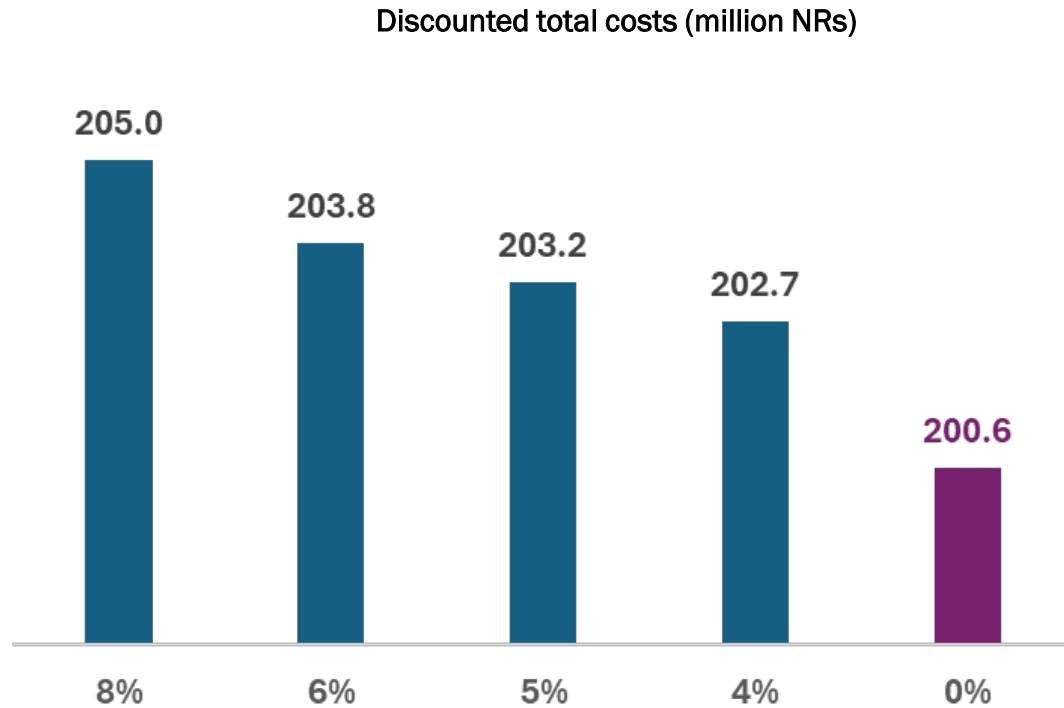
Total Levelized cost (Base case)

Average annual emissions from boilers (1tph capacity) without pollution control equipment (PCE) (Base case)

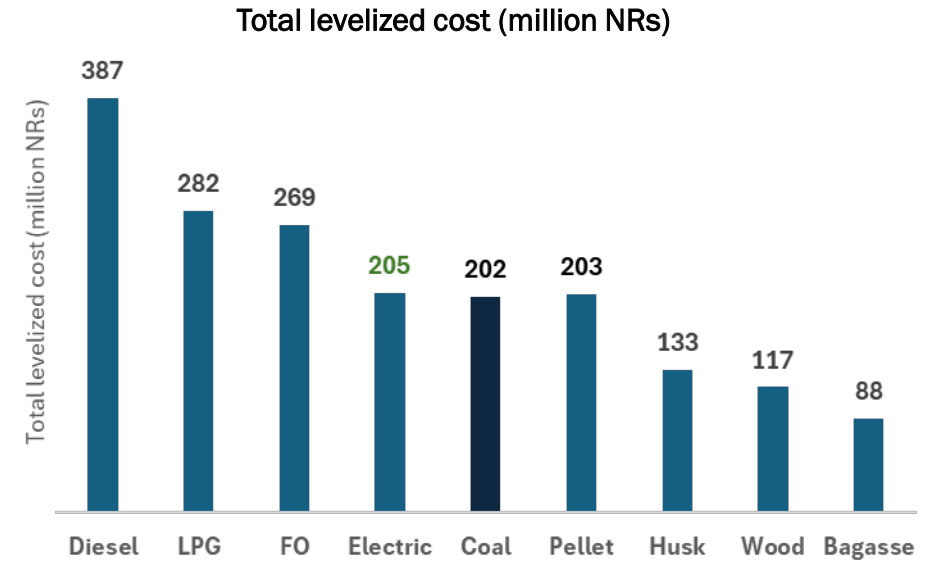


- PM2.5 is mostly from **biomass** and **coal**, and CO2 is from **fossil fuel** boilers.
- **Rice husk** contributes most to **PM2.5** and **Coal** contributes most to **CO2**.
- **Electric** boilers have zero emissions, assuming it is hydro-based..

Total levelized cost (Electric) with Lower Interest Rate



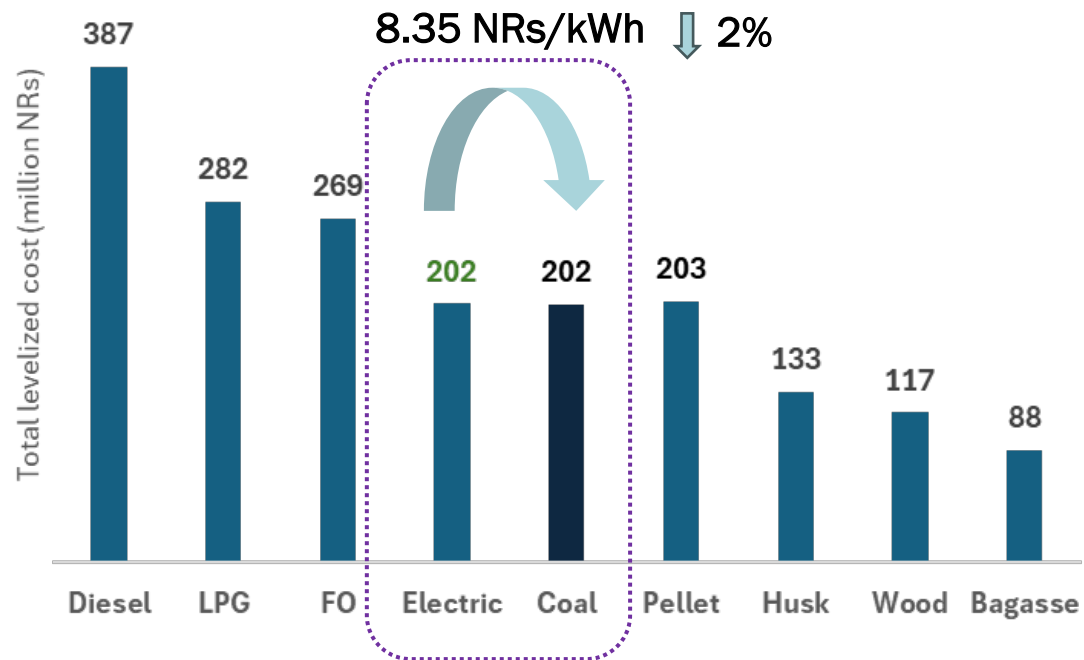
Base case



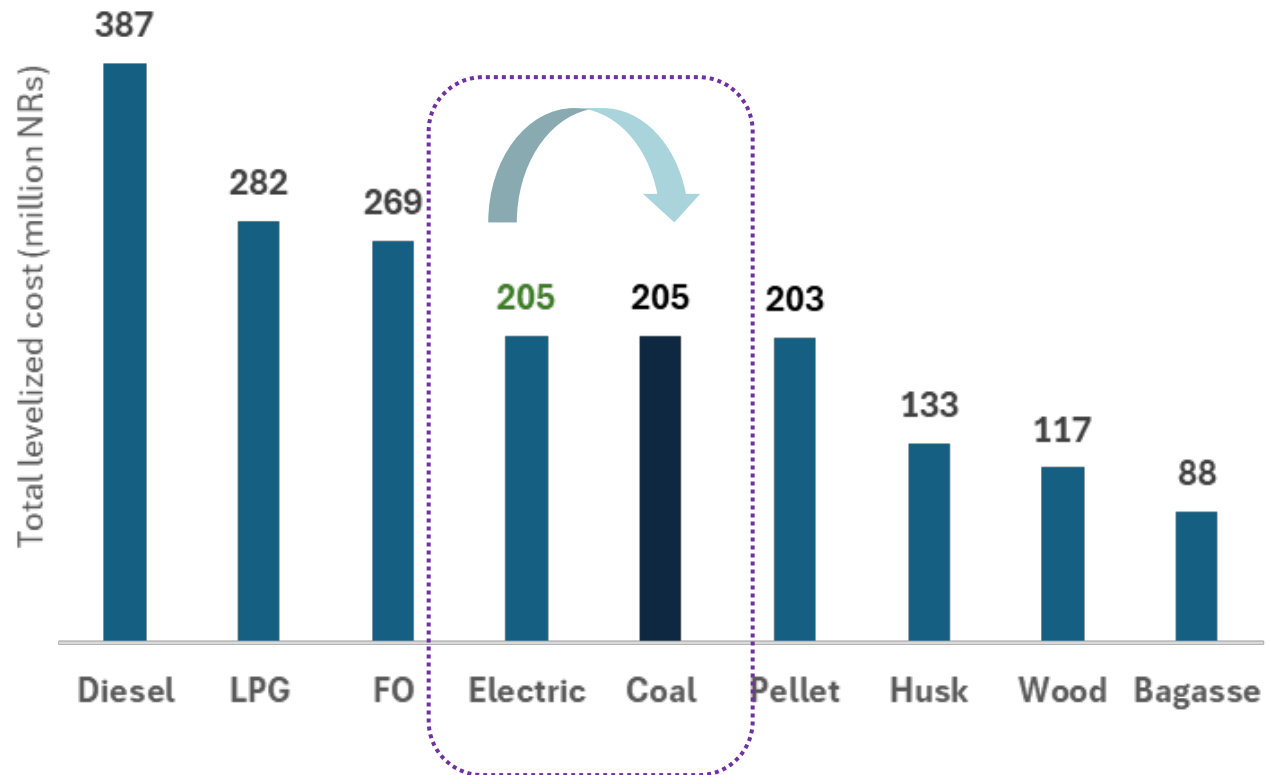
Even with **zero-interest** loans, electric boilers still rank as the fourth most expensive option.

Impact of capital subsidy

Production loan to make the electric boiler attractive with fossil fuel (Coal)



Increase in the price of fossil fuels (Coal) price

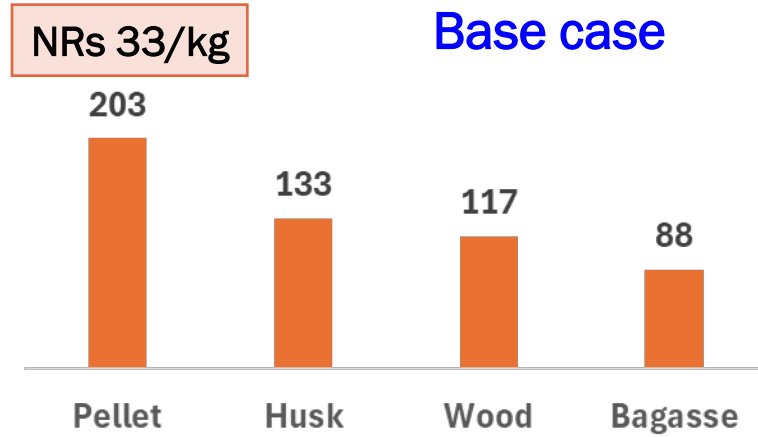
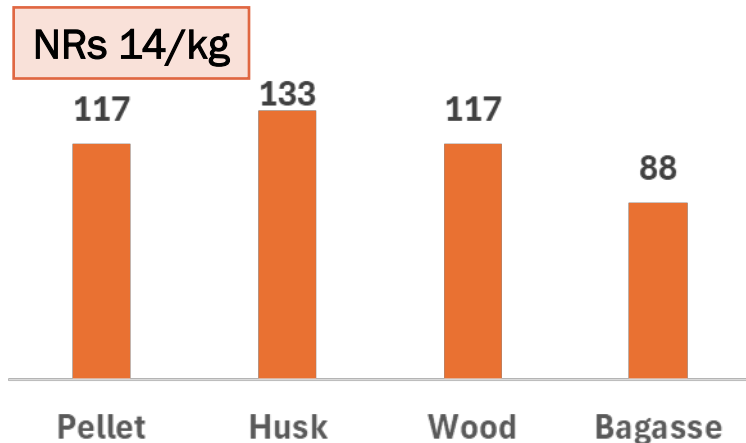
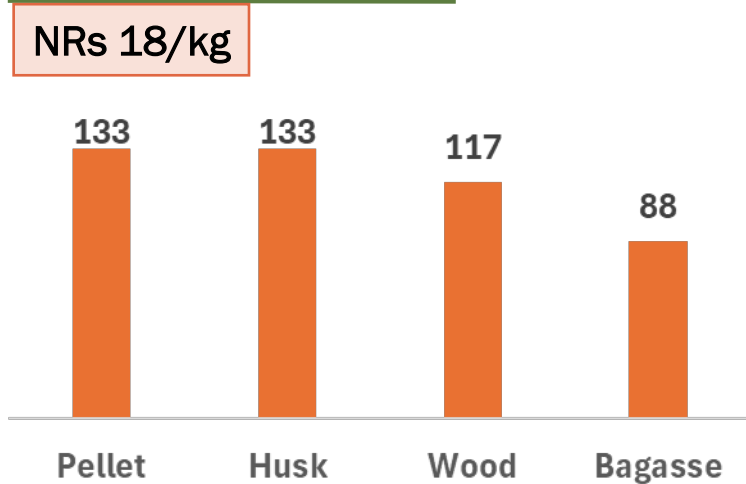


Fuel price (NRs/kg)

	Base case	New	% Ch
Coal	40	41	2.5%

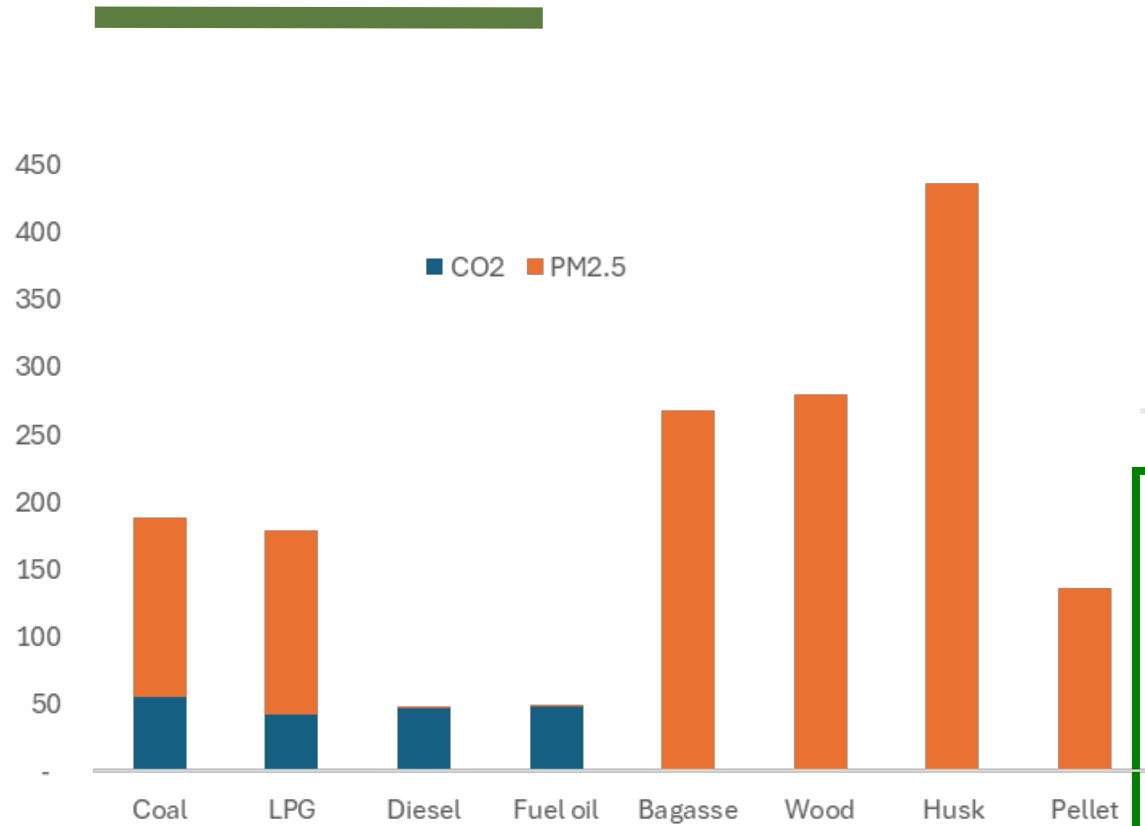
Increase in the fuel price of coal (carbon tax) to match with electricity.

Pellet promotion (million NRs)

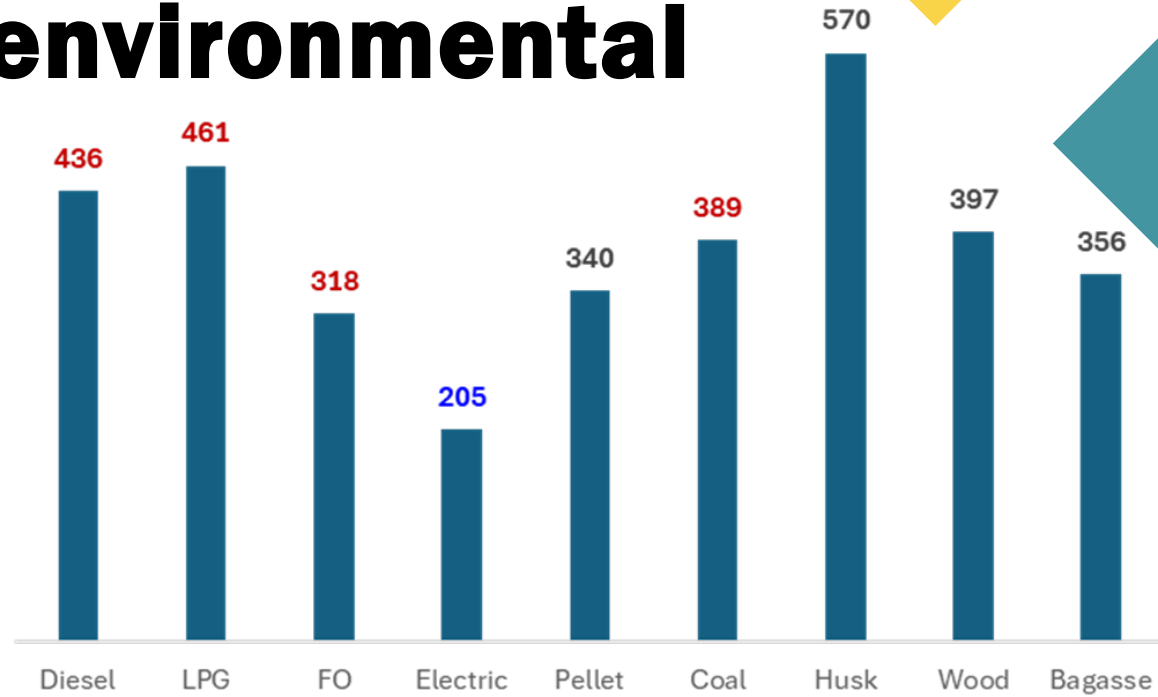


- **46% (58%)** price **reduction** from the current price to make the **pellet** boiler competitive with **husk (wood)**.
- Production costs of pellets vary between 9.2 to 19.6 NRs/kg, depending on the feedstock used, with 14.1 NRs/kg for wood chips-based pellet production.
- This also **incentivizes** the promotion of **domestic pelletization**.

Total annualized with environmental costs (million NRs)



Note: The social costs of CO2 is USD 0.04/kg of CO2, and PM2.5 is USD 86.85/kg of PM2.5. Source: EPIC India (2018).



- Environmental damage costs
 - Environmental damage cost (CO2)
 - Health damage cost (PM2.5)
- With these costs, **Electric** boilers are most **economical**
- **Biomass** boilers are relatively more **expensive** compared to **fossil fuel** boilers.

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Conclusions

- **Capital subsidies** (e.g., lower interest rates) are not much help. Even with **zero-interest** loans, **electric** boilers still rank **fourth** most expensive (after diesel, LPG, and FO and before coal).
- **Fuel costs are the most significant** (39% to 86% of the total annualized cost).
- **Production loan/subsidy** (lowering electricity price):
 - For the **electric boiler** to be competitive with **coal**, **2% decrease** in electricity price is needed.
- A **46% (58%) reduction** in the **price** of **pellets** makes **pellet** boiler **competitive** with **husk (wood)**.
- Attractive **financially** (suppliers) and **cheaper** than fossil fuel boilers (industries), yet pellet production in Nepal has **not** been scaled up.
- If we account for **environmental costs**, an **electric** boiler becomes the **cheapest** option.



Thank you

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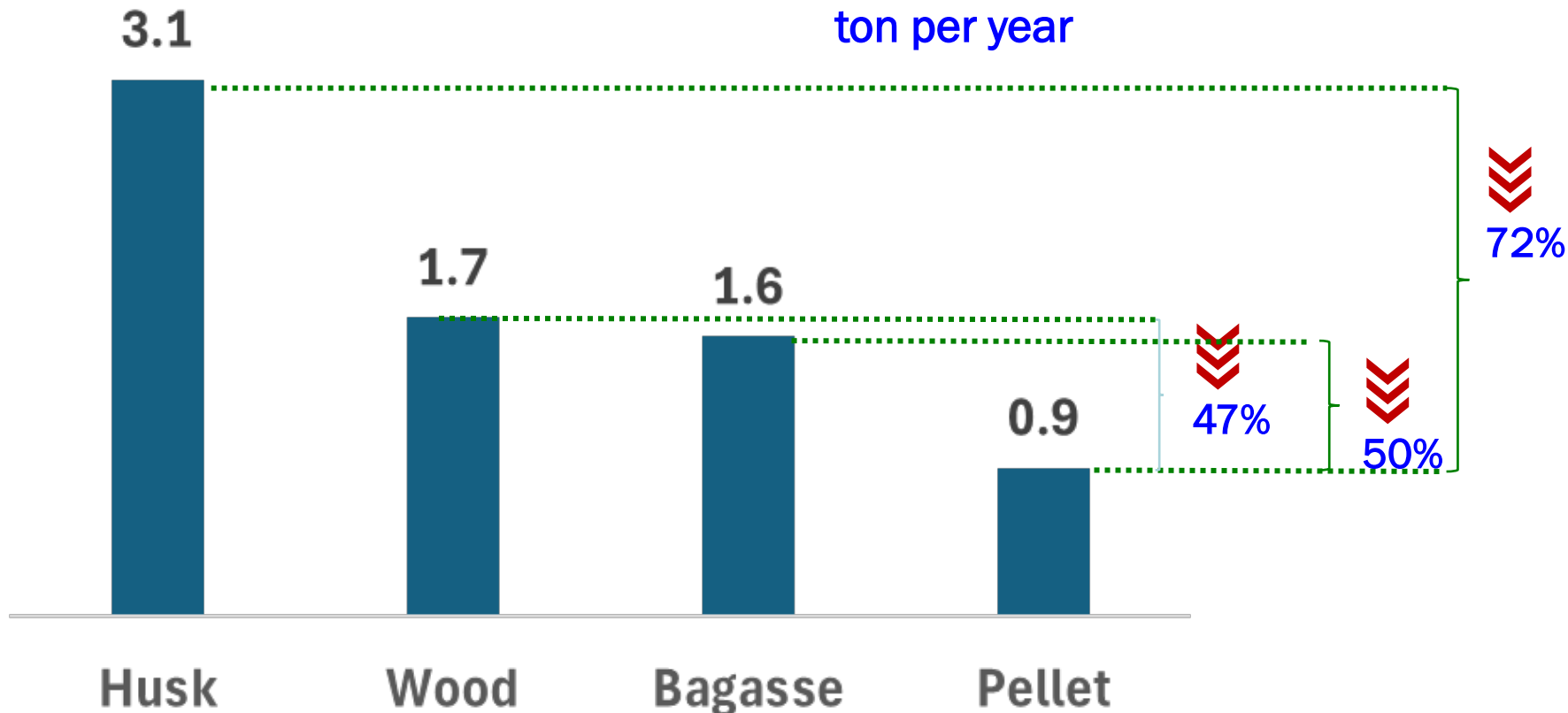
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Extra Slides

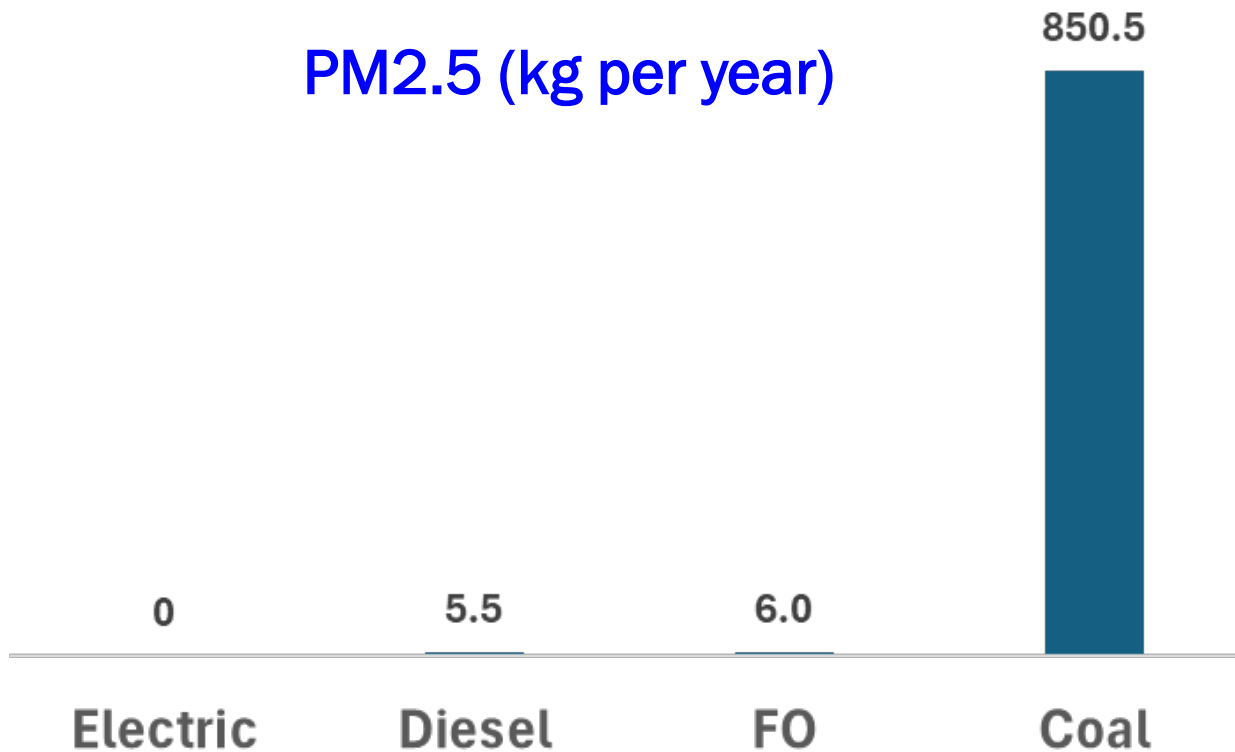


Average PM2.5 emissions of biomass boilers (1 tph capacity)



- PM2.5 emission is much **lower** with **pellets**.
- About **47% lower** than **bagasse**, **50% lower** than **fuelwood** and **72% lower** than **husk** boilers.

Average PM2.5 emissions of fossil-fuel boilers (1 tph capacity)



Although the **levelized cost of coal** boilers is cheaper than electric boilers, the **PM2.5** emissions associated with coal boilers are **very high**.